



API DOCUMENTATION

VIRTUAL ACCOUNT SERVICE

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OVERVIEW

This document describes endpoints for Virtual Account Service offered by the bank. Services are:

1. Authentication
2. Virtual account creation.
3. Reallocate virtual account.
4. Deactivate virtual account.
5. Reactivate virtual account.
6. Confirm virtual account payment.
7. Refund virtual account payment.
8. Notification.

AUTHENTICATION AND AUTHORIZATION

DESCRIPTION: All APIs require a Bearer token for authorization. Client is provided with a public key and private key for authentication and these credentials are used to generate an Authorization Bearer token with the endpoint below. Generated token must be passed in header of all other API requests.

URL: **POST** <https://baastest.9psb.com.ng/iva-api/v1/merchant/virtualaccount/authenticate>

REQUEST BODY:

Field	Description	Data Type	Mandatory
publickey	Client public key	String	Y
privatekey	Client private key	String	Y

RESPONSE:

Field	Description	Data Type
status	Response status	String
code	Response code	String
message	Response message	String
access_token	Authorization Bearer token	String
expires_in	Token expiry time in seconds	Int

SAMPLE REQUEST AND RESPONSE

Request

```
{
  "publickey": "",
  "privatekey": ""
}

Success response

{
  "access_token":
    "eyJhbGciOiJIUzUxMiIsInR5cCI6IkpXVCJ9.eyJ1bmVlYWQiaWQiOiIOTVBMTY0NkQzMjc0QTZEZkVCMENEQzJDMTBDREUyQ2SiImh0dHA6Ly9yZWZlbnhtbHNvYXAub3JnL3dzLzlwMDUvMDUvaWRlbmRpdHkvY2xhaWw1ZL2hhc2giOiJjNjc0MWMwM2NjU2YmM4NmUwMDk0OGE0ZGI1ZTRkYTFiZWUxM2RiYjNjMjEzZjkwZTU4YjE5NWl0YTAwZDE3NGU2NjA1NTc0OGRIYTcyMjg0YTEyMTAzYjRkZjVkZmRmNjQwMDdhNjhiYzYwYjBhMWQ5M2U4NTg4YzQ5NWlyYyslnJvbGUiaWUiOiJpd3piUWUxM1NDRI9rVDBNQjV5cnU1NVpMOEZacnFUYkppM2JsK2EyNnFwT283MVpwNSt1VjAxZmxBVUpXUmptVVVySDJtY1BVazNld3h1UXY3YlFCbkq2bjRCbGloUittNctIR2ZvejBzUjFicVdUZ1lqd3Q4WjJpSHRNtkNhUWwQMzZybGhKUHDETVJSYnlPejA2RzV1TzdOLzhYzJxWnp5bWJJYTkrcSTd6K3JRclJpY1hpQjJaSmJDL1FhcVICZGtzRXIPMNn1cG5qc1hjNVcyb3k3bmdNcUFjeXNaQjd6c3NFdnFsK0tvWstWNELheKx6MDdyN2E1Y25MelhvYjZaa2hmUkVJTG1sTnlPdkJav2xiVUpta0ILQU5GSTNZNU9RdU80Z3dSRVg3Z1h2K2ITTEk1S0ViOWxKWVpmRGNg6aTRMem9qUnZjdVpTNVNfcUJ4QXIYNIV1dUxhSXZlcDEyNEpSQnl1bzIGNUJWeW5CVWIwZ3Z6QVMrUVUilCJuYmYoijE2ODY0ODM4OTkslmV4cCl6MTY4NjQ5MTA5OSwiaWF0IjojoxNjg2NDgzODk5LCJpc3MiOiI1UfNCX1NlcnZpY2VfSW50ZXJuYWwiLCJhdWQiOiI1UfNCX1NlcnZpY2VfSW50ZXJuYWwifQ.0mdBaNIoiM_RTeCMUv3oukkpNcmCGotyGa0VJzxV8EVBADsH4uvV5c11LACl2Vr6Tm83uxc3xJHxsAAkb4JdsoQ"
```

```

    "expires_in": 7200,
    "code": "00",
    "message": "Success"
  }

```

Failure response

```

{
  "access_token": null,
  "expires_in": 0,
  "code": "S1",
  "message": "Invalid Credential"
}

```

VIRTUAL ACCOUNT CREATION

DESCRIPTION: Virtual account creations are of two type- STATIC and DYNAMIC. The following rules below applied to virtual account creation:

- i. When account.type is DYNAMIC, the expiry tag is mandatory.
- ii. "hours" field takes a minimum of 1 and maximum of 24.
- iii. Amount can be omitted if amounttype is "ANY". iv. "Amounttype" field is mandatory, and value can be:
 - a. EXACT: which means customer must pay the exact amount, otherwise the transaction will not be accepted.
 - b. ANY: which means customer can pay any amount, not just the amount specified in the order tag. With ANY, amount tag can be zero.
 - c. HIGHEROREXACT: which means customer must pay at least the amount specified in the order tag or higher, but not lower.
 - d. LOWEROREXACT: which means customer must pay at most the amount specified in the order tag or lower, but not higher.
- v. beneficiarytocredit tag is optional. Only necessary where a direct/instant credit is required to the ultimate beneficiary account. The account number and bank code need to be supplied. For 9PSB account, bank code is 120001.

URL: **POST** <https://baastest.9psb.com.ng/iva-api/v1/merchant/virtualaccount/create>

REQUEST BODY:

Field	Description	Data type	Mandatory
transaction	JSON Object of transaction	JSON Object	Y
transaction.reference	Transaction reference for creating virtual account	String	Y
order	JSON Object of order	JSON Object	Y
order.amount	amount	Double	Y
order.currency	currency	String	Y
order.description	description	String	

order.country	country	String	Y
order.amounttype	Amount type	String	Y
customer	JSON Object of customer	JSON Object	Y
customer.account	JSON Object of customer's account	JSON Object	Y
customer.account.name	Customer's name	String	Y
customer.account.type	Virtual account's type	String	Y
beneficiarytocredit.accountnumber	Ultimate beneficiary account to credit	String	N
beneficiarytocredit.bankcode	The bank code (six digit NIBSS bank code)	String	N

RESPONSE BODY:

Field	Description	Data Type
code	Response code	String
message	Response message	String
transaction	JSON Object of transaction	JSON Object
transaction.reference	Transaction reference for creating virtual account	String
order	JSON Object of order	JSON Object
order.amount	amount	Double
order.currency	currency	String
order.description	description	String
order.country	country	String
order.amounttype	Amount type	String
customer	JSON Object of customer	JSON Object
customer.account	JSON Object of customer's account	JSON Object
customer.account.name	Customer's name	String
customer.account.type	Virtual account's type	String
customer.account.expiry	JSON Object of customer's account expiry hours and date	JSON Object
customer.account.expiry.hours	Expiry's hour	int
customer.account.expiry.date	Expiry's date	Timestamp
customer.account.number	Account number	String
customer.account.bank	bank	String
beneficiarytocredit.accountnumber	Ultimate beneficiary account to credit	String
beneficiarytocredit.bankcode	The bank code (six digit NIBSS bank code)	String

beneficiarytocredit.feeamount	Charges to be deducted from the payment amount and credited to partner's account.	double
-------------------------------	---	--------

SAMPLE REQUEST AND RESPONSE

Request

i. Static example

```
{
  "transaction": {
    "reference": "20230606083659123456200"
  },
  "order": {
    "amount": 100,
    "currency": "NGN",
    "description": "Test TRF",
    "country": "NGA",
    "amounttype": "EXACT"
  },
  "customer": {
    "account": {
      "name": "Ayinla Azeez",
      "type": "STATIC"
    }
  }
}
```

ii. Dynamic example

```
{
  "code": "00",
  "message": "Success",
  "transaction": {
    "reference": "20230606083659123456300"
  },
  "order": {
    "amount": 100,
    "currency": "NGN",
    "description": "Test TRF",
    "country": "NGA",
    "amounttype": "EXACT"
  },
  "customer": {
    "account": {
      "name": "Bolaji Oreagba",
      "type": "DYNAMIC",
      "expiry": {
        "hours": 1,
        "date": "2023-06-11T13:47:31.7993952+01:00"
      }
    }
  },
}
```

```

        "number": "5030000013",
        "bank": "9PSB"
    }
},

    "beneficiarytocredit": {
        "accountnumber": "1100000299",
        "bankcode": "120001",
        "feeamount": 5.50
    }
}

```

Success response

i. Static

```

{
    "code": "00",
    "message": "Success",
    "transaction": {
        "reference": "20230606083659123456200"
    },
    "order": {
        "amount": 100,
        "currency": "NGN",
        "description": "Test TRF",
        "country": "NGA",
        "amounttype": "EXACT"
    },
    "customer": {
        "account": {
            "name": "Ayinla Azeez",
            "type": "STATIC",
            "expiry": null,
            "number": "5030000116",
            "bank": "9PSB"
        }
    }
}

```

ii. Dynamic

```

{
    "code": "00",
    "message": "Success",
    "transaction": {
        "reference": "20230606083659123456300"
    },
    "order": {
        "amount": 100,
        "currency": "NGN",

```

```

        "description": "Test TRF",
        "country": "NGA",
        "amounttype": "EXACT"
    },
    "customer": {
        "account": {
            "name": "Bolaji Oreagba",
            "type": "DYNAMIC",
            "expiry": {
                "hours": 1,
                "date": "2023-06-11T13:47:31.7993952+01:00"
            },
        },
        "number": "5030000013",
        "bank": "9PSB"
    }
},

    "beneficiarytocredit": {
        "accountnumber": "1100000299",
        "bankcode": "120001",
        "feeamount": 5.50
    }

}

```

Failure response

```

{
    "code": "47",
    "message": "Invalid transaction request - reference number must be unique.",
    "transaction": {
        "reference": "20230606083659123456300"
    },
    "order": {
        "amount": 100,
        "currency": "NGN",
        "description": "Test TRF",
        "country": "NGA",
        "amounttype": "EXACT"
    },
    "customer": {
        "account": {
            "name": "Bolaji Oreagba",
            "type": "DYNAMIC",
            "expiry": {
                "hours": 1,
                "date": null
            },
        },
        "number": null,
        "bank": null
    }
}

```

```
}
```

For invalid credential, response below:

```
{  
  "code": "S1",  
  "message": "Invalid Credential"  
}
```

REALLOCATE VIRTUAL ACCOUNT

DESCRIPTION: This service initiate reallocation of virtual account. The following points below must be considered when reallocating virtual account:

1. the “reference” field is a new unique reference, not the reference used to create the original virtual account to be reallocated.
2. the rest of the parameter is the same as when creating dynamic or virtual account.
3. the “number” field is mandatory and it virtual account number to be reallocated.
4. any virtual account can be reallocated.

URL: **POST** <https://baatest.9psb.com.ng/iva-api/v1/merchant/virtualaccount/reallocate>

REQUEST BODY:

Field	Description	Data type	Mandatory
transaction	JSON Object of transaction	JSON Object	Y
transaction.reference	Transaction reference for creating virtual account	String	Y
order	JSON Object of order	JSON Object	Y
order.amount	amount	Double	Y
order.currency	currency	String	Y
order.description	description	String	
order.country	country	String	Y
order.amounttype	Amount type	String	Y
customer	JSON Object of customer	JSON Object	Y
customer.account	JSON Object of customer's account	JSON Object	Y
customer.account.name	Customer's name	String	Y
customer.account.type	Virtual account's type	String	Y
Field	Description	Data type	Mandatory
customer.account.number	account number	String	Y

customer.account.expiry	JSON Object of customer's account expiry time	JSON Object	Y
customer.account.expiry.hours	Account expiry time in hours	int	Y
beneficiarytocredit.accountnumber	Ultimate beneficiary account to credit	String	N
beneficiarytocredit.bankcode	The bank code (six digit NIBSS bank code)	String	N

RESPONSE BODY:

Field	Description	Data Type
code	Response code	String
message	Response message	String
transaction.reference	Transaction reference for creating virtual account	JSON Object
order	JSON Object of order	JSON Object
order.amount	amount	Double
order.currency	currency	String
order.description	description	String
order.country	country	String
order.amounttype	Amount type	String
customer	JSON Object of customer	JSON Object
customer.account	JSON Object of customer's account	JSON Object
customer.account.name	Customer's name	String
customer.account.type	Virtual account's type	String
customer.account.expiry	JSON Object of customer's account expiry hours and date	JSON Object
customer.account.expiry.hours	Expiry's hour	int
customer.account.expiry.date	Expiry's date	Timestamp
customer.account.number	Account number	String
customer.account.bank	bank	String
beneficiarytocredit.accountnumber	Ultimate beneficiary account to credit	String
beneficiarytocredit.bankcode	The bank code (six digit NIBSS bank code)	String
beneficiarytocredit.feeamount	Charges to be deducted from the payment amount and credited to partner's account.	double

SAMPLE REQUEST AND RESPONSE

Request

```
{
  "transaction": {
    "reference": "202306051414559123261"
  },
  "order": {
    "amount": 1500,
    "currency": "NGN",
    "description": "Test TRF",
    "country": "NGA",
    "amounttype": "ANY"
  },
  "customer": {
    "account": {
      "name": "Olufunke Rhodalene",
      "number": "5030000075",
      "type": "DYNAMIC",
      "expiry": {
        "hours": 12
      }
    }
  },
  "beneficiarytocredit": {
    "accountnumber": "1100000299",
    "bankcode": "120001",
    "feeamount": 5.50
  }
}
```

Success response

```
{
  "code": "00",
  "message": "Success",
  "transaction": {
    "reference": "202306051414559123261"
  },
  "order": {
    "amount": 1500,
    "currency": "NGN",
    "description": "Test TRF",
    "country": "NGA",
    "amounttype": "ANY"
  },
  "customer": {
    "account": {
      "name": "Olufunke Rhodalene",
      "type": "DYNAMIC",
      "expiry": {
        "hours": 12,

```

```

        "date": "2023-06-12T00:49:00.4689189+01:00"
      },
      "number": "5030000075",
      "bank": "9PSB"
    }
  },
  "beneficiarytocredit": {
    "accountnumber": "1100000299",
    "bankcode": "120001",
    "feeamount": 5.50
  }
}

```

Failure response

```

{
  "code": "47",
  "message": "Invalid transaction request - reference number must be unique.",
  "transaction": {
    "reference": "202306051414559123261"
  },
  "order": {
    "amount": 1500,
    "currency": "NGN",
    "description": "Test TRF",
    "country": "NGA",
    "amounttype": "ANY"
  },
  "customer": {
    "account": {
      "name": "Olufunke Rhodalene",
      "type": "DYNAMIC",
      "expiry": {
        "hours": 12,
        "date": null
      },
    },
    "number": "5030000075",
    "bank": null
  }
}

```

For invalid credential, response below:

```

{
  "code": "S1",
  "message": "Invalid Credential"
}

```

DEACTIVATE VIRTUAL ACCOUNT

DESCRIPTION: This service deactivates existing virtual account. The following points below must be considered when deactivating virtual account:

1. the “reference” field is mandatory can be original reference used to create or last used reallocate the virtual account, or a new unique one, the original reference used to create the virtual account will not be changed on the system.
2. the “number” field is mandatory and its the virtual account number to deactivate.

URL: **POST** <https://baatest.9psb.com.ng/iva-api/v1/merchant/virtualaccount/deactivate>

REQUEST BODY:

Field	Description	Data type	Mandatory
transaction	JSON Object of transaction	JSON Object	Y
transaction.reference	Transaction reference for creating virtual account	String	Y
customer	JSON Object of customer	JSON Object	Y
customer.account	JSON Object of customer's account	JSON Object	Y
customer.account.number	account number	String	Y

RESPONSE BODY:

Field	Description	Data Type
code	Response code	String
message	Response message	String
transaction	JSON Object of transaction	JSON Object
transaction.reference	Transaction reference for creating virtual account	String
customer	JSON Object of customer	JSON Object
customer.account	JSON Object of customer's account	JSON Object
customer.account.name	Customer's name	String
customer.account.type	Virtual account's type	String
customer.account.expiry	customer's account expiry time	String
customer.account.number	Account number	String
customer.account.bank	bank	String

SAMPLE REQUEST AND RESPONSE

Request

```
{
```

```

"transaction": {
  "reference": "20230606093659123456101"
},
"customer": {
  "account": {
    "number": "5030000051"
  }
}
}

```

Success response

```

{
  "code": "00",
  "message": "Success",
  "transaction": {
    "reference": "20230606093659123456101"
  }, "customer": {
    "account": {
      "name": "9PSB Agent/OLAIDE ABASS",
      "type": "STATIC",
      "expiry": null,
      "number": "5030000051",
      "bank": "9PSB"
    }
  }
}

```

Failure response

```

{
  "code": "S1",
  "message": "Invalid Credential"
}

```

REACTIVATE VIRTUAL ACCOUNT

DESCRIPTION: This service deactivates existing virtual account. The following points below must be considered when deactivating virtual account:

1. the “reference” field is mandatory can be original reference used to create or last used reallocate the virtual account, or a new unique one, the original reference used to create the virtual account will not be changed on the system.
2. the “number” field is mandatory and its the virtual account number to reactivate.
3. the virtual account goes back to active with exact setting it had before deactivation.

URL: **POST** <https://baatest.9psb.com.ng/iva-api/v1/merchant/virtualaccount/reactivate>

REQUEST BODY:

Field	Description	Data type	Mandatory
transaction	JSON Object of transaction	JSON Object	Y
transaction.reference	Transaction reference for creating virtual account	String	Y
customer	JSON Object of customer	JSON Object	Y
customer.account	JSON Object of customer's account	JSON Object	Y
customer.account.number	account number	String	Y

RESPONSE BODY:

Field	Description	Data Type
code	Response code	String
message	Response message	String
transaction	JSON Object of transaction	JSON Object
transaction.reference	Transaction reference for creating virtual account	String
customer	JSON Object of customer	JSON Object
customer.account	JSON Object of customer's account	JSON Object
customer.account.name	Customer's name	String
customer.account.type	Virtual account's type	String
customer.account.expiry	customer's account expiry time	String
customer.account.number	Account number	String
customer.account.bank	bank	String

SAMPLE REQUEST AND RESPONSE**Request**

```
{
  "transaction": {
    "reference": "20230606093659123456100"
  },
  "customer": {
    "account": {
      "number": "5030000051"
    }
  }
}
```

Success response

```
{
  "code": "00",
  "message": "Success",
  "transaction": {
    "reference": "20230606093659123456100"
  },
  "customer": {
    "account": {
      "name": "9PSB Agent/OLAIDE ABASS",
      "type": "STATIC",
      "expiry": null,
      "number": "5030000051",
      "bank": "9PSB"
    }
  }
}
```

Failure response

```
{
  "code": "47",
  "message": "Invalid transaction request - account number to deactivate not found.",
  "transaction": {
    "reference": "20230606093659123456100"
  },
  "customer": {
    "account": {
      "name": null,
      "type": null,
      "expiry": null,
      "number": "503000005177777",
      "bank": null
    }
  }
}
```

CONFIRM VIRTUAL ACCOUNT PAYMENT

DESCRIPTION: This service is used to confirm virtual account payment. The following points below must be considered when confirming virtual account:

1. the “accountnumber” field is mandatory.
2. The “reference” field is the virtual account creation,”sessionid” field is the transaction session id if customer provide it. One must be provided, the two can be provided, but the two cannot be blank.
3. the “amount” field is not compulsory, but when provided transaction with that exact amount will be returned if found.
4. the maximum of latest ten records that matches the parameter combination supplied will be returned.

URL: **POST** <https://baatest.9psb.com.ng/iva-api/v1/merchant/virtualaccount/confirmpayment>

REQUEST BODY:

Field	Description	Data type	Mandatory
reference	Transaction reference to confirm payment	String	Y if sessionid is N or N if sessionid is Y
sessionid	Transaction reference id	String	Y if reference is N or N if reference is Y
amount	Transaction amount	Double	N
accountnumber	account number	String	Y

RESPONSE BODY:

Field	Description	Data Type
code	Response code	String
message	Response message	String
transactions	JSON Array of transactions	JSON Array
transactions.transaction	JSON Object of transaction	JSON Object
transactions.transaction.reference	Transaction reference	String
transactions.transaction.sessionid	Transaction session id	String
transactions.transaction.date	Date of confirming payment	Timestamp
transactions.order	JSON Object of order	JSON Object
transactions.order.amount	amount	Double
transactions.order.currency	currency	String
transactions.order.description	description	String
transactions.customer	JSON Object of customer	JSON Object
transactions.customer.account	JSON Object of customer's account	JSON Object
transactions.customer.account.name	Customer's name	String
transactions.customer.account.senderbankname	Sender bank name	String
transactions.customer.account.senderbankcode	Sender bank code	String
transactions.customer.account.senderaccountnumber	Sender account number	String
transactions.customer.account.number	Customer ccount number	String
transactions.customer.account.bank	Sender bank	String

transactions.customer.account.sendername	Sender name	String
--	-------------	--------

SAMPLE REQUEST AND RESPONSE

Request

```
{
  "reference": "20230606083659123456102",
  "sessionid": "",
  "amount": 120,
  "accountnumber": "5030000013"
}
```

Success response

```
{
  "transactions": [
    {
      "transaction": {
        "reference": "20230606083659123456102",
        "sessionid": "999169230607093526326281438501",
        "date": "2023-06-07T09:35:29"
      },
      "customer": {
        "account": {
          "name": "9PSB Agent/Ajadi Taofeek",
          "number": "5030000013",
          "bank": "9PSB",
          "senderbankcode": "120001",
          "senderbankname": "9PSB",
          "senderaccountnumber": "1100000024",
          "sendername": "9PSB Agent"
        }
      },
      "order": {
        "amount": 120,
        "currency": "NGN",
        "description": "9PSB Agent/1100000024/TRF TEST NIP WITH NA/9PSB Agent-Ajadi
Tao/9PSB230607093526297062701"
      },
      "message": "Success",
      "code": "00"
    }
  ],
  "code": "00",
  "message": "Success"
}
```

Failure response

```
{
  "transactions": null,
  "code": "47",
}
```

```
"message": "Invalid transaction request - reference or sessionid is mandatory"
}
```

REFUND VIRTUAL ACCOUNT PAYMENT

DESCRIPTION: This service is used to refund full or part of virtual account payment. The following points below must be considered when using refund payment virtual account:

1. All request parameters field are mandatory.
2. The amount must less than (partial refund is supported) or equal the original payment.
3. The refund can be made only once for a payment, once a successful refund is made, even if its lower than the original amount, no further refund is possible for that payment.
4. Refund maximum period is seven days from payment.
5. Refund Hash is SHA512 string composed as follows:

`privateKey + sessionid + accountnumber + amount + DateOfSubmission`

Sample for validation

Parameters: 9IWxL3OfOC4GO6JIQan4qlx9X8wBcz8YVjd1xizN_HR4gdlsrW0Mf7EUDQt5d1tC + 999169231016134100460496227401+ 5030045748 + 100.00 + 20231024

Combined: :

9IWxL3OfOC4GO6JIQan4qlx9X8wBcz8YVjd1xizN_HR4gdlsrW0Mf7EUDQt5d1tC9991692310161341004604962274015030045748100.0020231024

Hash:

9AAB82A465201CCFE0C22A047DDCF41CFFE9D686C37415FF671613EDE6B2EFB023DE6BE682E36C8D2CCB1F613EA9DA688E6028C7C5DD98CCACC985BF0965175A

Note: Amount is in 2 decimal places and DateOfSubmission is in format yyyyMMdd.

URL: **POST** <https://baatest.9psb.com.ng/iva-api/v1/merchant/virtualaccount/paymentrefund>

REQUEST BODY:

Field	Description	Data type	Mandatory
sessionid	Transaction session id as contained in the payment notification or confirm payment payload	String	Y
hash	Request hash as detailed in the description section	String	Y
amount	amount to refund, must be less than or	Double	Y

	equal the original payment amount		
accountnumber	Virtual account number	String	Y

RESPONSE BODY:

Field	Description	Data Type
code	Response code	String
message	Response message	String
transaction.reference	Transaction reference	String
transaction.linkingreference	Transaction internal reference	String
transaction.externalreference	Transaction session id	String
transaction.date	Date of confirming payment	Timestamp
order.amount	amount	Double
order.currency	currency (NGN)	String
order.description	description	String
order.country	country (NGA)	String
customer	JSON Object of customer's account	JSON Object
customer.account	JSON Object of customer's account	JSON Object
customer.account.name	Sender Customer's name	String
customer.account.kyc	Customer KYC	String
customer.account.senderbankcode	Sender bank code	String
customer.account.senderaccountnumber	Sender account number (merchant account number debited for refund)	String
customer.account.number	Customer account number	String
customer.account.bvn	BVN (where available)	String
customer.account.bank	Sender bank	String
customer.account.sendername	Sender name	String

SAMPLE REQUEST AND RESPONSE

Request

```
{
```

```
"sessionId":"999169231016134100460496227401",
"amount":"100.00",
"accountnumber":"5030045748",
"hash":"9AAB82A465201CCFE0C22A047DDCF41CFFE9D686C37415FF671613EDE6B2EFB023DE6BE682E36C8D2
CCB1F613EA9DA688E6028C7C5DD98CCACC985BF0965175A"
}
```

Success response

```
{
  "transaction": {
    "reference": "999169231016134100460496227401R",
    "linkingreference": "VT2310141041158328768758172793",
    "externalreference": "999169231024134100460496221234",
    "date": "2023-10-24T11:53:25.8424917+01:00"
  },
  "order": {
    "amount": 100.0,
    "description": "VTS-REFUND-999169231016134100460496227401",
    "currency": "NGN",
    "country": "NGA"
  },
  "customer": {
    "account": {
      "number": "1100000309",
      "bank": "120001",
      "name": "RENZE HAVEMAN",
      "bvn": "2222222222",
      "senderaccountnumber": "1100015371",
      "sendername": "Merchant Name",
      "kyc": null
    }
  },
  "code": "00",
  "message": "Approved Or Completed Successfully"
}
```

Failure response

```
{
  "transaction": null,
  "order": null,
  "customer": null,
  "code": "47",
  "message": "Invalid transaction request - original payment not found"
}
```

```
{
```

```

"transaction": null,
"order": null,
"customer": null,
"code": "74",
"message": "Invalid Message Signature"
}

```

NOTIFICATION

DESCRIPTION: Partner provides the url where any inflow will be sent into their virtual account.

1. Basic authentication is supported for the notification service only.
2. The system will retry repush five times if response code 00 is not returned for the notification, and after 5 times, system will stop.
3. Hash is required to be validated by the partner to further strengthen notification security and message integrity. Hash is SHA512 string composed as follows (password is the credential password provided to us for the basic authentication):

password + senderAccountNumber + SenderBankCode + virtualAccountNumber + TransactionAmount

Sample for validation

Parameters: hdUof9b92D0WS40HnOhgtsem + 1003245245 + 000008 + 5030000013 + 76.70

Combined: hdUof9b92D0WS40HnOhgtsem1003245245000008503000001376.70

Hash:

8938CFEF854D1953A89CFD8F77EA4E0E8B2700D67ED949D6278EEB5735066D1EC4274F430
F2231CD6F324BF5A2E7A43AFCBBF9A93E9F9BD778A83D28EECAE6BC

Note: TransactionAmount is in 2 decimal places, Hash will come in caps.

4. We require that our IPs be whitelisted on your environment for call security in addition to the basic authentication and message hash.
5. Once you got our notification, clients are advised to call our **confirmpayment** endpoint before applying fund to the account/customer position on their side, especially and mandatorily if our IPs were not whitelisted. The request should be:

```

{
  "sessionId": "20230606083659123456102",
  "amount": 120,
  "accountnumber": "5030000013"
}

```

This will always return one record because session id unique.

EXPECTED REQUEST BODY:

Field	Description	Data Type
code	Response code	String
message	Response message	String
transaction	JSON Object of transaction	JSON Object
hash	Message signature	String
transaction.reference	Transaction reference	String
transaction.sessionid	Transaction session id	String
transaction.date	Date of confirming payment	Timestamp
order	JSON Object of order	JSON Object
order.amount	amount	Double
order.currency	currency	String
order.description	description	String
customer	JSON Object of customer	JSON Object
customer.account	JSON Object of customer's account	JSON Object
customer.account.name	Customer's name	String
customer.account.senderbankname	Sender bank name	String
customer.account.senderbankcode	Sender bank code	String
customer.account.senderaccountnumber	Sender account number	String
customer.account.number	Customer account number	String
customer.account.bank	Sender's bank	String
customer.account.sendername	Sender's name	String

EXPECTED RESPONSE BODY:

Field	Description	Data Type
code	Response code	String
message	Response message	String

SAMPLE REQUEST AND RESPONSE

Request

```
{
  "transaction": {
    "reference": "20230606083659123456102",
    "sessionid": "999169230607093526326281438501",
    "date": "2023-06-07T09:35:29"
  },
  "customer": {
    "account": {
      "name": "9PSB Agent/Ajadi Taofeek",
      "number": "5030000013",

```

```

    "bank": "9PSB",
    "senderbankcode": "120001",
    "senderbankname": "9PSB",
    "senderaccountnumber": "1100000024",
    "sendername": "Benson Malik"
  }
},
"order": {
  "amount": 120.0,
  "currency": "NGN",
  "description": "9PSBAgent/1100000024/TRF TEST NIP WITH NA/9PSB Agent-Ajadi
Tao/9PSB230607093526297062701"
},
"message": "Success",
"code": "00",
"hash": "ABD12343EF5678"
}
Response
{
  "message": "Success",
  "code": "00"
}

```

RESPONSE CODES

Response Code	Message	Description
S1	Invalid credential	When invalidation authentication is passed or call is not authorized due to IP issue or method called not permitted.
00	Successful	Operation successful
47	Invalidation transaction request	The detail issue will be part of the message, e.g. where account type is invalid, i.e. neither STATIC not DYNAMIC
99	Request processing error	System error occurred, such transaction can be re-tried, and if issue persist, please escalate to the bank support.